

Federated Learning for Human Activity Recognition with Environmental and Subject-Level Awareness

Berke Cansiz**

berke.cansiz@std.yildiz.edu.tr

Department of Electronics and Communication
Engineering, Yildiz Technical University
Istanbul, Türkiye

Murat Taskiran

mrttskrn@yildiz.edu.tr

Department of Electronics and Communication
Engineering, Yildiz Technical University
Istanbul, Türkiye

Hatice Vildan Dudukcu

vdudukcu@yildiz.edu.tr

Department of Electronics and Communication
Engineering, Yildiz Technical University
Istanbul, Türkiye

Nihan Kahraman

nicoskun@yildiz.edu.tr

Department of Electronics and Communication
Engineering, Yildiz Technical University
Istanbul, Türkiye

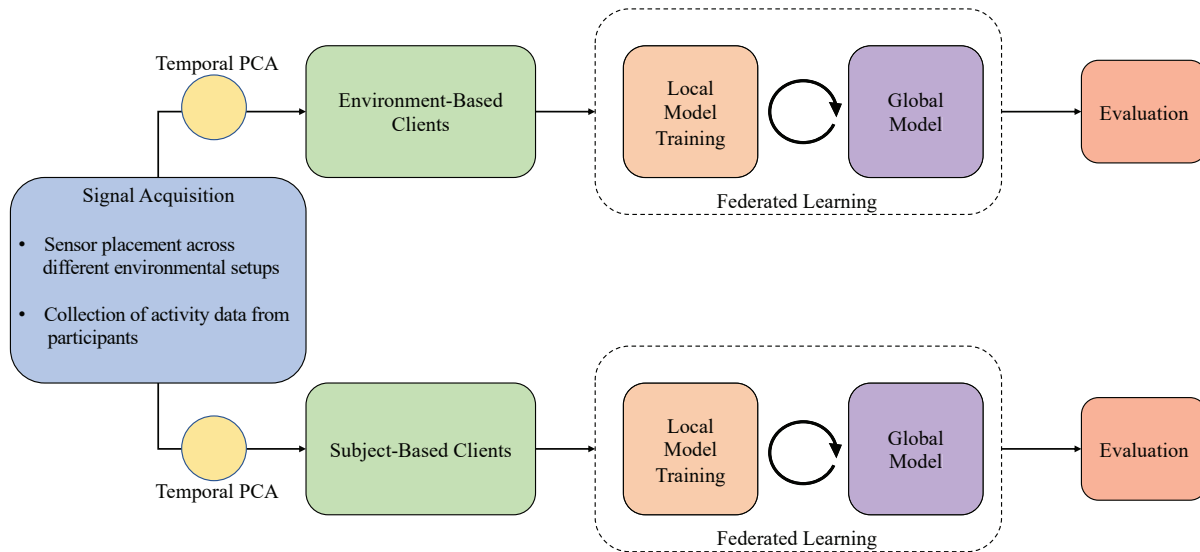


Figure 1: The Block Diagram of the Proposed Method

Abstract

In human activity recognition studies, the location of the sensors and environmental conditions are determined with precision during the data collection process, and this positioning can directly affect the model performance. In addition, the individual characteristics of the participants while performing the activities stand out as a separate problem element that makes the generalizability of the system difficult. In order to overcome these two problems, the federated learning method was applied in the presented study to obtain a model that adapts to different environmental conditions and

has a high generalization capacity among users. Within the scope of this study, studies were carried out on two different scenarios. In these scenarios, client types were determined as environment-based and subject-based and the studies were carried out in this direction. In addition, the effect of aggregation functions on system performance was examined in these two scenarios. The experimental results show that FedAdam achieves superior performance in environment-based systems, while FedYogi achieves superior performance in subject-based systems.

Keywords

Human Activity Recognition, Federated Learning, Aggregation Functions, Machine Learning

ACM Reference Format:

Berke Cansiz, Murat Taskiran, Hatice Vildan Dudukcu, and Nihan Kahraman. 2025. Federated Learning for Human Activity Recognition with Environmental and Subject-Level Awareness. In *Companion of the the 2025*

*Both authors contributed equally to this research.



ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25), October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3756149>

1 Introduction

Human Activity Recognition (HAR) has emerged as a key element in many intelligent systems, such as home automation, health-care monitoring, and human-computer interaction. Conventional Human Activity Recognition methods depend significantly on labeled data acquired in controlled environments with static sensor configurations. However, when exposed to unknown situations or new participants, real-world deployments frequently suffer from differences in user-specific traits and contextual contexts, which significantly impair model performance. The consistency of the input data can be significantly impacted by the positioning of the sensor and ambient environmental conditions, such as temperature, illumination, or surface type. At the same time, it is challenging to create models that effectively generalize among users due to individual variations in behavior, body dynamics, and activity executions [4, 12]. These challenges hinder the scalability and reliability of centralized Human Activity Recognition systems, particularly where privacy and personalization are essential. With the use of federated learning (FL), a potential decentralized methodology, many clients (such as environments or users) may work together to train machine learning models without exchanging raw data. This approach is especially well-suited for HAR applications, where the distribution of data is inherently non-independent and identically distributed and data privacy and personalization are important.

Recently, federated learning (FL) [7] has been widely used in the field of human activity recognition (HAR) to achieve high accuracy while preserving privacy. A review study by Aouedi and Sacco comprehensively discusses how FL-based HAR systems increase both data privacy and computational efficiency [2]. In the proposed method by Ndeko et al., the “Federated Split Learning + Differential Privacy (FSL-DP)” system was created and it was aimed to provide both speed and accuracy gains with the created system [8]. The model, which uses both accelerometer and gyroscope data, has been shown by simulation studies to achieve better accuracy with lower communication overhead compared to classical FL. In the study conducted by Albogamy, a FL framework implemented with hybrid LSTM-GRU models running on IoMT devices was proposed. It was tested on UCI-HAR, HARTH and HAR7+ datasets. In the proposed system, 1DCNN convolution blocks were combined with LSTM-GRU based time series analysis layers and attention mechanism was used to increase the recognition accuracy. In addition, weighted Federated Averaging (FedAvg) was applied to solve the data heterogeneity and unbalanced distribution problems. The results obtained using UCI-HAR, HARTH and HAR7+ datasets showed that it provides performance close to centralized training while preserving privacy [1]. The LN-FHAR method proposed by Sun et al. proposes a two-stage federated learning (FL) approach to solve the label noise problem frequently encountered in sensor-based human activity recognition (HAR) systems. In the first stage, the label noise on the client devices is detected and cleaned, while in the second stage, central model training is performed with the cleaned data. Thus, both privacy is protected and model corruptions caused by label errors are prevented. In the simulation studies, it

has been shown that the proposed method provides higher accuracy compared to traditional FL methods on data containing noisy labels [11]. When the FL solutions used in the literature to solve the Human Activity Recognition problem are examined, it is clearly seen that they mainly focus on increasing privacy, increasing performance and reducing communication load.

The present study proposes a federated learning-based human activity recognition framework that addresses two primary limitations: generalization across diverse environmental circumstances and robustness between various participants while maintaining user privacy. Two federated scenarios are established: one where local models are built for each setting and another where models are trained for each participant. In both scenarios, a generalized global model is created by combining model updates on a central server. This study assess how well federated learning improves HAR generalizability by utilizing a four-layer Multilayer Perceptron (MLP) architecture and contrasting four well-known federated optimization strategies: FedAvg [7], FedProx[13], FedYogi [9], and FedAdam [9]. Our results show that federated learning can make it possible to create scalable, privacy-preserving HAR systems that function consistently in a variety of real-world scenarios.

In the following sections of the paper, in Section 2, detailed information is given about the method proposed within the scope of the study and the dataset and model architecture used are explained in detail and information is given about experimental setup. In Section 3 the obtained results are shared and these results are interpreted. In the last section, the outputs obtained within the scope of this study are presented and suggestions are made for future studies.

2 Methodology

The implemented framework includes two distinct federated learning scenarios: one based on environmental variation and the other based on subject diversity. An overview of the system architecture is presented in the block diagram in Figure 1.

2.1 Dataset

This study employs the ESP32 Dataset[5, 6], collected within an indoor area. Data were collected under three indoor experimental setups: two Line-of-Sight (LOS) and one Non-Line-of-Sight (NLOS) scenario. These setups were designed to examine the impact of particularly proximity and obstruction on the propagation of radio frequency (RF) signals and the model performance. In LOS Scenario 1, the Tx and Rx were placed 7 meters apart, and participants performed activities at the midpoint between them. Similarly, in LOS Scenario 2 again with the activity zone positioned at the midpoint the Tx–Rx distance was changed to 5.2 meters. In the NLOS Scenario, a wooden wall was placed between the Tx and Rx to simulate signal obstruction.

The participants were 8 volunteers (3 females, 5 males) and they were instructed to perform all activities at the midpoint between Tx and Rx to ensure balanced signal strength. Each participant performed 10 predefined activities given in Table 1, with 20 trials per activity, resulting in 1,600 recordings per scenario.

Table 1: ESP32 Dataset Activities

Activity Indicator	Activity
A01	Jumping jack
A02	Squatting
A03	Hand swing (front and back)
A04	Walking from the centre towards the Rx
A05	Walking from the centre towards the Tx
A06	Bouncing basketball
A07	Jogging in place
A08	Forward bend 90°
A09	Sitting down on chair
A10	Standing

2.2 Model Architecture and Implementation

In this study, to classify human activities based on wireless signal features, a fully connected four-layer feedforward neural network was implemented using PyTorch. The model receives an input vector including 104 features, which denote signal-based attributes derived from each activity sample. At each time step, a 104-dimensional feature vector was acquired for each sample. Principal Component Analysis (PCA) was employed along the temporal axis to provide a fixed-length representation of the sample, suitable for the model. This produced a single 1×104 feature vector for each sample by reducing the temporal information. The architecture includes three hidden layers with decreasing dimensionality of 256, 128, and 64 neurons, respectively.

Each hidden layer includes a linear transformation followed by layer normalization, a rectified linear unit (ReLU) activation function, and a dropout layer with a dropout rate of 0.1. The use of layer normalization helps stabilize and accelerate training [3], while dropout serves as a regularization technique that reduces overfitting [10]. Whereas ReLU activations introduce non-linearity into the model, enabling it to capture complex feature interactions.

The final layer is a fully connected output layer that maps the last hidden representation to a ten-dimensional output vector, corresponding to the ten activity classes in the dataset. The entire architecture is lightweight and computationally efficient, making it suitable for deployment in decentralized environments where federated learning is applied.

FL essentially consists of two main components: the client side and the server side. Model training is performed on the client side on local datasets, while the server side performs the aggregation of updated model parameters [7]. In this study, the specified model architectures have been deployed on the client side, and four different aggregation methods are applied on the server side. These methods are FedAvg [7], which averages the model parameters; FedProx [13], which uses coefficient-based computation to reduce divergence in local training; and two optimizer-based momentum methods, FedAdam and FedYogi [9], which leverage adaptive optimization techniques to improve convergence.

2.3 Evaluation Metrics

Accuracy is one of the basic metrics used to measure the accuracy of a classification model. It is defined as the ratio of the number of

correctly predicted examples by the model to the total number of examples. It is a reliable performance indicator, especially when the classes are balanced. However, it can be misleading when the classes are unbalanced. The Accuracy metric is mathematically expressed as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

The F1 score is used to evaluate the performance of the model in a more balanced way. It is the harmonic average of the Precision and Recall metrics and is especially preferred in unbalanced data sets. In this way, both the ability of the model to make true positive predictions and its ability to find true positives are evaluated together. The F1 score is calculated with the following formula:

$$\text{F1 Score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \quad (2)$$

The calculation of Precision and Recall mentioned in this equation is as in the following equations:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

All metrics are calculated based on true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) derived from the confusion matrix.

2.4 Experimental Setup

Two separate experimental setups, one based on environment-based federated learning and the other on user-based federated learning, were developed to evaluate the proposed federated learning framework for human activity recognition. Both configurations aim to evaluate the generalizability of the trained models over diverse settings and client variability, reflecting real-world deployment situations.

In the environment-based federated scenario, each of the three indoor configurations, which corresponded to two LOS and one NLOS environment, was handled as its own federated client. Each client had access only to the data that was collected in their own physical environment. This included all of the volunteers and types of activities that were recorded in that area. In this setup, local models were trained separately in each environment utilizing only the data from that environment. After local training epochs, model weights were sent to a central server, where global aggregation was done using federated optimization algorithms including FedAvg, FedProx, FedYogi, and FedAdam. This scenario was designed to assess whether a globally shared model could generalize across different environmental conditions and physical configurations. On the other hand, the user-based federated scenario handled each volunteer as a distinct federated client. In this case, the dataset was split up by participant, and each client trained a local model on their own activity trials in all of the environmental configurations. This method puts data privacy first and is in line with real-world situations when personal sensor data cannot be held in one place because of legal or moral reasons. The purpose of this scenario was to create a model that could work for people with diverse physical

traits and behavior patterns without needing access to user-specific data during global training.

Both experimental configurations employed previously described model architecture and training hyperparameters. Local training for each client was performed over 5 epochs, with uniform hyperparameters for learning rate (0.001), batch size (64), and optimizer configurations (Adam optimizer). Evaluation metrics include classification accuracy, F1-score. These two experimental methods offer different perspectives into model robustness: one concerning environmental variability and the other about user diversity. Collectively, they constitute a thorough assessment of federated learning for radio frequency-based human activity identification in practical and privacy-conscious settings.

3 Results and Discussion

After the setup was completed as explained in the Experimental setup section, experimental studies were carried out using ESP32 dataset for two different client types and the results obtained in the experiments using different aggregation functions are shared in Table 2. In Table 2, both Accuracy and F1-score values are used to examine the system performance. The reason for this is that the accuracy value alone is not sufficient to evaluate the system in imbalanced data. In addition, the results with the best performance are given in bold in Table 2. When the results in Table 2 are examined, it is seen that the environment-based client type achieves higher performance than the subject-based client type in the proposed system. In addition, when the results are examined in terms of aggregation functions, it is clear that the FedAdam function is more successful in the environment-based client type and the FedYogi function is more successful in the subject-based client type.

Table 2: Accuracy and F1-scores obtained for Different Client Types

	Client Type			
	Environment-Based		Subject-Based	
Algorithm	Accuracy	F1-Score	Accuracy	F1-Score
FedAvg [7]	0.8479	0.8466	0.8208	0.8198
FedProx [13]	0.8552	0.8544	0.8135	0.8125
FedYogi [9]	0.8510	0.8501	0.8552	0.8535
FedAdam [9]	0.8635	0.8621	0.8375	0.8361

In addition to the information given in Table 2, in order to examine how the aggregation functions used in the proposed system converge, performance change graphs were plotted according to rounds for two different client types and these graphs are given in Figure 2 and Figure 3, respectively. When the graphs in Figure 2 and Figure 3 are examined in detail, it is clearly seen that the convergence speed of the FedYogi function is much better than the other functions. In the FedAdam function, the low convergence speed, especially in the first rounds, is noteworthy.

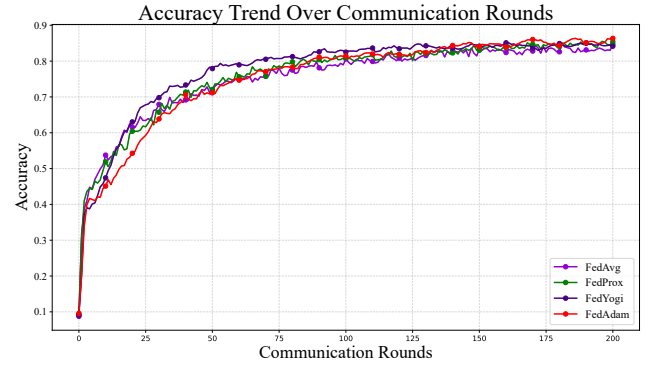


Figure 2: Accuracy Change Graph Depending on Communication Rounds. (Environment-Based System)

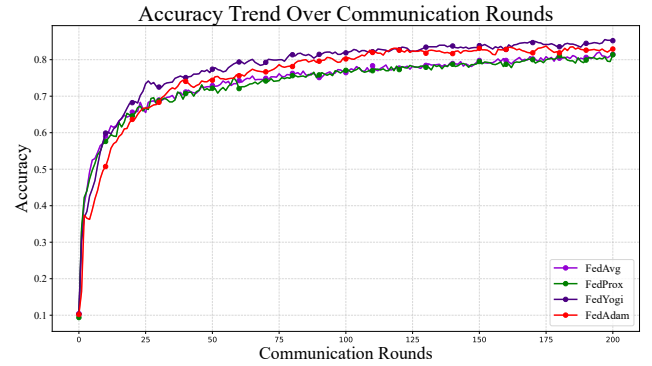


Figure 3: Accuracy Change Graph Depending on Communication Rounds. (Subject-Based System)

4 Conclusion and Future Works

In this study, which proposes a different perspective on federated learning approaches in solving the human activity recognition problem, which has become increasingly popular among researchers in recent years, the performances and convergence rates of different aggregation functions for two different client types were examined and shared. When the results obtained in the experimental studies carried out using the ESP32 Dataset are examined, although different aggregation functions gave better results in different client types, when a total examination was performed, it was observed that the FedYogi function was better than the other functions in terms of performance. In addition, the changes in performance according to the communication rounds were also examined within the scope of the study and it was observed that FedYogi was the function with the best convergence rate in the proposed system.

In future studies, it is aimed to increase the client types and to propose different aggregation functions to produce new solutions. Especially in recent times, ensemble-based aggregation functions are becoming increasingly widespread and it is thought that a new approach to be proposed on this basis will be positive for the performance of the system.

5 Acknowledgments

This work was supported and funded by the European Union (Project No. 101071179). The views and opinions expressed are those of the authors only and do not necessarily reflect those of the European Union or EISMEA. Neither the European Union nor the granting authority can be held responsible for them.

References

- [1] Fahad R Albogamy. 2025. Federated Learning for IoMT-Enhanced Human Activity Recognition with Hybrid LSTM-GRU Networks. *Sensors* 25, 3 (2025), 907.
- [2] Ons Aouedi, Alessio Sacco, Latif U Khan, Dinh C Nguyen, and Mohsen Guizani. 2024. Federated Learning for Human Activity Recognition: Overview, Advances, and Challenges. *IEEE Open Journal of the Communications Society* (2024).
- [3] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. 2016. Layer normalization. *arXiv preprint arXiv:1607.06450* (2016).
- [4] Yang Li, Guanci Yang, Zhidong Su, Shaobo Li, and Yang Wang. 2023. Human activity recognition based on multienvironment sensor data. *Information Fusion* 91 (2023), 47–63.
- [5] Zhe-Yu Lim and Lee-Yeng Ong. 2024. Radio Frequency-based Human Activity Dataset collected using the ESP32 Microcontroller in Line-of-sight and Non-line-of-sight Indoor Experiment Setups. (2024). doi:10.17632/x4x5xtvtwt.1
- [6] Zhe-Yu Lim, Lee-Yeng Ong, and Meng-Chew Leow. 2024. Radio frequency-based human activity dataset collected using ESP32 microcontroller in line-of-sight and non-line-of-sight indoor experiment setups. *Data in Brief* 57 (2024), 111101.
- [7] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. PMLR, 1273–1282.
- [8] Josue Ndeko, Shaba Shaon, Aubrey Beal, Avimanyu Sahoo, and Dinh C Nguyen. 2024. Federated Split Learning for Human Activity Recognition with Differential Privacy. *arXiv preprint arXiv:2411.06263* (2024).
- [9] Sashank Reddi, Zachary Charles, Manzil Zaheer, Zachary Garrett, Keith Rush, Jakub Konečný, Sanjiv Kumar, and H Brendan McMahan. 2020. Adaptive federated optimization. *arXiv preprint arXiv:2003.00295* (2020).
- [10] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research* 15, 56 (2014), 1929–1958. <http://jmlr.org/papers/v15/srivastava14a.html>
- [11] Haifeng Sun, Junping Yao, Xiaojun Li, Yanfei Liu, and Hongyang Gu. 2025. Robust two stages federated learning for sensor based human activity recognition with label noise. *Scientific Reports* 15, 1 (2025), 1–16.
- [12] Santosh Kumar Yadav, Kamlesh Tiwari, Hari Mohan Pandey, and Shaik Ali Akbar. 2021. A review of multimodal human activity recognition with special emphasis on classification, applications, challenges and future directions. *Knowledge-Based Systems* 223 (2021), 106970.
- [13] Dong Yin, Yudong Chen, Ramchandran Kannan, and Peter Bartlett. 2018. Byzantine-robust distributed learning: Towards optimal statistical rates. In *International conference on machine learning*. Pmlr, 5650–5659.